

LAITH ASSAF

East Lansing, MI | (248) 480-5783 | assaflai@msu.edu | [linkedin.com/in/laith-assaf-](https://www.linkedin.com/in/laith-assaf-) | github.com/Novapool

SUMMARY

Computer Science student who designs systems and then explains them. Architected a multi-service game platform and a real-estate pricing model, and spent three years translating technical failures into plain language for non-technical users.

EDUCATION

Michigan State University, College of Engineering

East Lansing, MI

BS Computer Science, AI/ML | Minor in Economics | Dean's List x3

August 2022 - May 2027

- **Coursework:** Cloud Computing & Database Systems, Computer Systems, Data Structures & Algorithms

EXPERIENCE

PC Technician Intern

May 2026 - Present

Onsite Support Team, Delta Dental

Okemos, MI

- Deployed and configured **150+** workstations across a new corporate office buildout, imaging endpoints via **Ivanti** MDM to enforce one standardized configuration across the fleet rather than per-machine setup

IT and Media Services Support Associate

May 2023 - Present

MSU College of Osteopathic Medicine

East Lansing, MI

- Supported 100+ classroom technology systems across 4 locations serving **1,200+ students**, resolving **50-70 tickets weekly** and communicating fixes to faculty with no technical background mid-lecture

PROJECTS

Bonfire Platform & Ember Framework | *TypeScript, Socket.io, Firebase*

December 2025 – April 2026

- Architected a **multi-room orchestration system** on Express + **Socket.io** namespaces with **Firebase** state sync, hosting 4 concurrent games on one **Railway** server with configurable disconnect and TTL cleanup
- Extracted the shared game lifecycle into **Ember**, a published open-source framework (3 npm packages), trading upfront abstraction cost for **80%+ code reuse** on every subsequent build

AI-Assisted Comparative Market Analysis | *Python, scikit-learn, Streamlit*

June 2026 – Present

- Built a sold-price prediction tool over metro Detroit **MLS** history reaching **13.3% MAPE**, surfacing the comparable sales behind every estimate so an agent can defend the number to a client
- Recommended **against** launch after benchmarking output at 13.3% MAPE versus the **5%** agent-grade bar, tracing the gap to missing source fields rather than model choice, and staged the roadmap around closing the data gap first

Toyota GR Cup Telemetry Analysis | *Python, PostgreSQL, scikit-learn*

2025

- Modeled tire degradation across **3,257 laps** from 7 racetracks, reaching **$R^2 = 0.631$** on high-frequency telemetry

Self-Hosted Home Lab Infrastructure | *Linux, Docker, Networking*

2023 – Present

- Administer a rack server (**128GB+ RAM**) running containerized media, game, and **VPN** services under **Docker**

LEADERSHIP

Michigan State University | AI Club

East Lansing, MI

Workshop Coordinator & Project Lead

January 2025 - Present

- Presented 8+ technical workshops on **local LLM deployment** and **PyTorch** fundamentals to 500+ attendees, and led a 4-student team scoping and shipping an **LLM-powered** academic predictor

TECHNICAL SKILLS

Architecture & Cloud: system design, distributed state sync, REST APIs, WebSockets, Docker, Railway, Vercel

Data: PostgreSQL, SQL, pandas, scikit-learn, ETL and preprocessing pipelines

Languages: Python, TypeScript, C++, C#, SQL

Platforms & Tools: Linux, Windows, Git, FastAPI, Flask, Express, Next.js, Active Directory, Intune, Ivanti MDM